$f(n) \in O(n^2)$ — ① (Proved)

T.S.T: $f(n) \in \Omega(n^2)$ i.e $f(n) \geq cn^2$

$f(n) = \frac{n^2}{2} - \frac{n}{2} \cdot 1 \geq \frac{n^2}{2} - \frac{n}{2} \cdot \frac{n}{2}$, take $n \geq 2$

$= \frac{n^2}{2} - \frac{n^2}{4}$

$= \frac{n^2}{4} = c(g(n))$

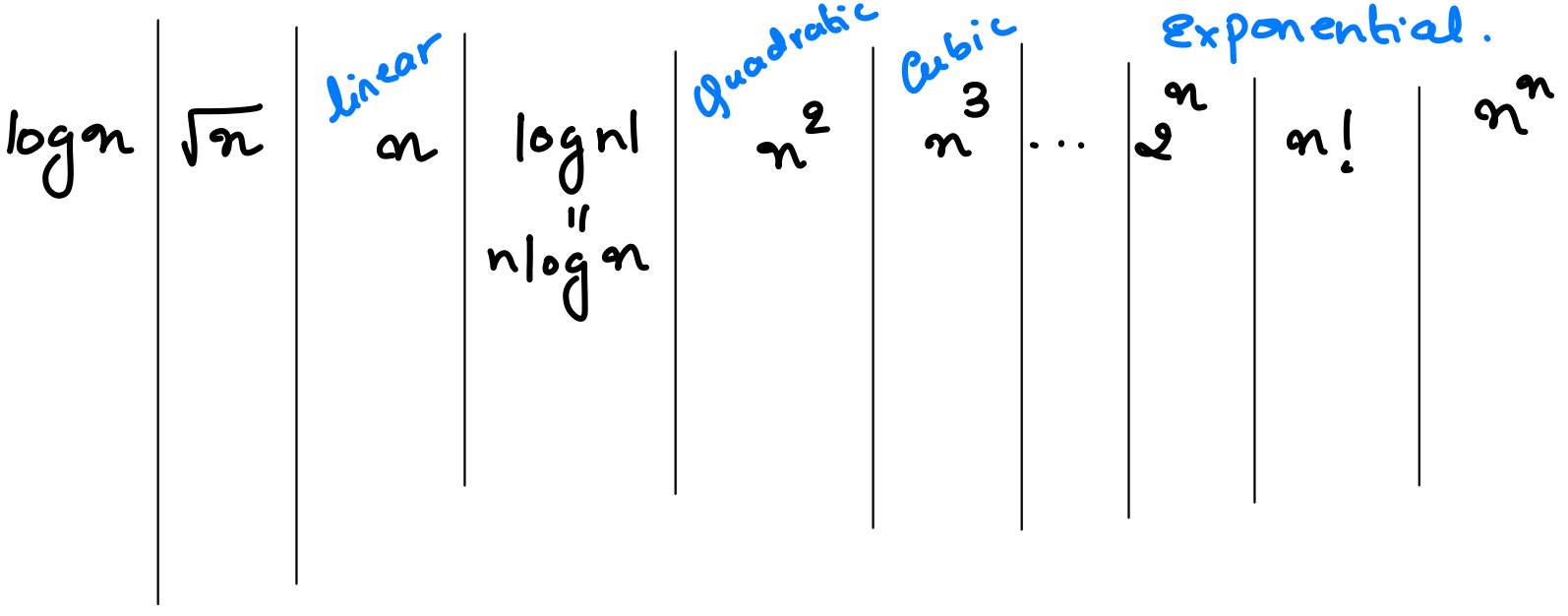
$\therefore f(n) \geq c\, g(n) \quad \forall \; n \geq n_0 = 2$ and $c = \frac{1}{4}$

$\Rightarrow f(n) \in \Omega(n^2)$ — (2)

$\therefore$ By (1) and (2), $f(n) \in \Theta(n^2)$

$\therefore$ Bubblesort Alg has quadratic growth.

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$\log n$
$\sqrt{n}$
linear
n
$n \log n$
$= n \log n$
Quadratic
n^2
Cubic
n^3
...
Exponential.
2^n
$n!$
n^n

$n^2 \leq n^3$

$\because \quad n^2 \cdot 1 \leq n^2 \cdot n = n^3$

$\sqrt{n} \leq n$

$\because \quad \sqrt{n} \cdot 1 \leq \sqrt{n} \cdot \sqrt{n} = n$

Q: $\log n \leq \sqrt{n}$?

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} \frac{n^2}{n^3} = \lim_{n\to\infty} \frac{1}{n} = 0.$$

$$\Rightarrow f(n) \leq g(n).$$

To find: $\lim_{n\to\infty} \frac{\log n}{\sqrt{n}}.$ $\left(\frac{\infty}{\infty}\right)$

$$\lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{1}{2}n^{-1/2}}$$

($\log_e$ Assumed)

$$\lim_{n\to\infty} 2 \times \frac{1}{n} \times \sqrt{n}$$

$$\lim_{n\to\infty} \frac{2}{\sqrt{n}} = 0.$$

$$\Rightarrow \log n \leq \sqrt{n}.$$

Recall: Tower of Hanoi - n disks

Steps involved: $2^n - 1$.

$$2^n \leq n^n \quad \text{for } n \geq 2.$$

Q: $2^n \leq n! \leq n^n$?

$$n! = n \times (n-1) \times \cdots \times 2 \times 1 \leq n \times n \times \cdots \times n = n^n$$

$\therefore \; n! \leq n^n.$

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T.S.T $2^n \leq n!$

Consider,

$$\lim_{n\to\infty} \frac{2^n}{n!} \qquad \left(\frac{\infty}{\infty}\right)$$

$$\lim_{x\to\infty} \frac{2^x}{x!} \qquad (x \in \mathbb{R})$$

$$\lim_{x\to\infty} \frac{x \log 2}{\log(x!)} \qquad -\circledast \qquad (\text{Applying } \log)$$

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Calculating $\log x!$:

Conts. def^n of $n!$
$\Gamma(n+1) = n!$

$$n! = 1 \cdot 2 \cdots n$$

$$\log n! = \log 1 + \log 2 + \cdots + \log n.$$

$$= (\log 1 + \log n) + (\log 2 + \log n-1) + \cdots + \left(\log\left[\frac{n}{2}\right] + \log\left[\frac{n}{2}\right]\right)$$

$$\geq \log \frac{n}{2} + \log \frac{n}{2} + \cdots + \log \frac{n}{2}$$

$$= \frac{n}{2}\log\frac{n}{2}$$

$$= \frac{n}{2}(\log n - \log 2)$$

$$= \frac{n}{4}\log n + \left(\frac{n}{4}\log n - \frac{n}{2}\log 2\right)$$

we want +ve

$$\geq \frac{n}{4}\log n, \quad n \geq n_0 = 4$$

$$c = 1/4$$

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$\Rightarrow \log n! \in \Omega(n \log n)$.

T.P. $\frac{n}{4}\log n \geq \frac{n}{2}\log 2$

$\frac{1}{4}\log n \geq \frac{1}{2}\log 2$

$\log n \geq 2\log_e 2$

After which n is this true?

$n \geq 4$.

$2\log_e 2 = 1.386294361$

n	$\log n$
1	0
2	0.69
3	1.09
4	1.386294361
5	1.609

$$n! \leq n^n$$

$$\log n! \leq n \log n$$

$$\Rightarrow \log n! \in O(n \log n)$$

$$\therefore \log n! \in \Theta(n \log n).$$

$$=$$

(*) $\Rightarrow$

$$\lim \frac{x \log 2}{\log x!} = \lim_{n \to \infty} \frac{n \log 2}{n \log n} = 0.$$

$$\therefore \quad 2^n \leq n!.$$

Another proof for $\log n! \in \Omega(n \log n)$

$$n! = 1 \cdot 2 \cdot 3 \cdots \left(\frac{n}{2}\right)\left(\frac{n}{2}+1\right) \cdots n$$

$$\geq \left(\frac{n}{2}+1\right)\left(\frac{n}{2}+2\right) \cdots n$$

$$\geq \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)\cdots\left(\frac{n}{2}\right)$$

$$= \left(\frac{n}{2}\right)^{n/2}$$

$$\log n! \geq \frac{n}{2}\log\frac{n}{2}$$

(Continuation as previous proof)

Calculating, $100! = (1 \cdot 2 \cdots 50)(51 \cdots 100)$ (Divide & Conquer)

$$= 50! \quad (51 \cdots 100)$$

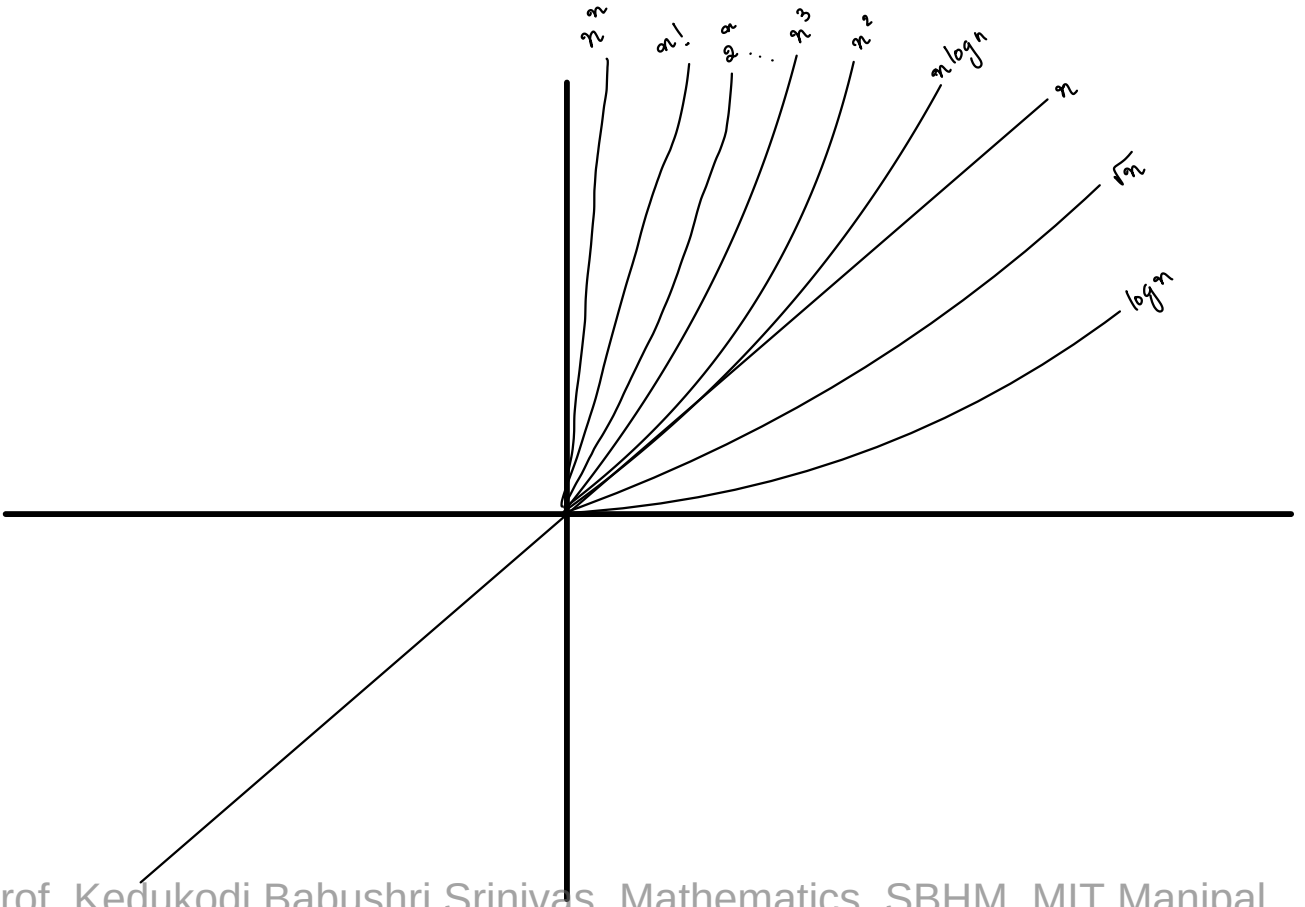
$$= (3.04 \times 10^{64})\left({}^{100}P_{50}\right)$$

$$\left({}^{11}_{3.06 \times 10^{93}}\right)$$

$$= 9.302 \times 10^{157}$$

	Sublinear		Linear	Polynomial				exponential		
n	$\log n$	$\sqrt{n}$	n	$n\log n$	n^2	n^3	...	2^n	$n!$	n^n
100	4.61	10	10^2	461	10^4	10^6		1.27×10^{30}	9.3×10^{157}	10^{200}

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2^n
$n!$
$2^n \cdots$
n^3
n^2
$n \log n$
n
$\sqrt{n}$
$\log n$

Computer → can compute 10^{12} operations per second

(Usual 1 core → 2.44 Hz ($\times 10^9$))

Given this computer, how much time does it take to solve Tower of Hanoi of 100 disks?

Steps : $2^{100} - 1$ (operations), Take 2^{100}

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$$\frac{2^{100}}{10^{12}} \text{ seconds}$$

1 sec : 10^{12}

? : 2^{100}

$$= \frac{2^{100}}{60 \times 10^{12}} \text{ minutes}$$

$$= \frac{1.27 \times 10^{30}}{60 \times 60 \times 10^{12}} \text{ hours}$$

$$= \frac{1.27 \times 10^{30}}{60 \times 60 \times 24 \times 10^{12}} \text{ days}$$

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$$= \frac{1.27 \times 10^{30}}{365 \times 60 \times 60 \times 24 \times 10^{12}} \text{ years}$$

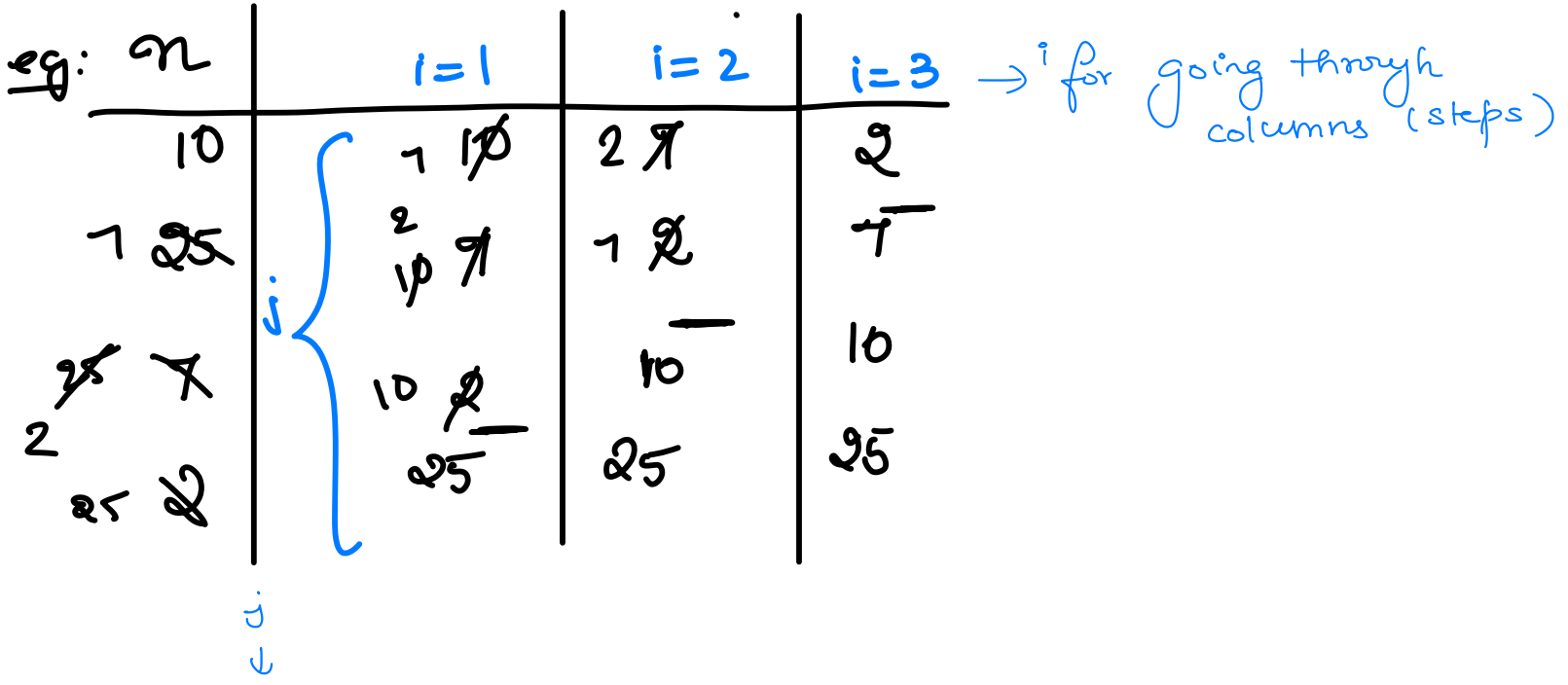
$$= 4.03 \times 10^{10} \text{ years.}$$

A computer which can compute 10^{12} operations per second takes 4.03×10^{10} years to solve Tower of Hanoi of 100 disks.

Age of Earth: 4.5×10^{9} years
4.5 Billion.

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Bubble sort Algorithm:



Input: Array of n elements $A[1, \cdots, n]$

No of steps involved : $n-1$

Algorithm:

For $i \leftarrow 1$ to $n-1$

  For $j \leftarrow 1$ to $n-i$

    If $A[j+1] < A[j]$, then $A[j+1] \leftrightarrow A[j]$

basic operation (comparision)

Verifications:

$i = 1$

$j = 1$ to $n-1$

$A[j+1]$

→ will consider $n-1+1 = n$.

$\therefore$ $j = 1$ to $n-i$ ✓

$A[j+1] < A[j]$ then $A[j+1] \leftrightarrow A[j]$

$25 < 10$ False ⇒ Nothing

$7 < 25$ True ⇒ swap ✓

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Calculating Time Complexity / No of comparisions:

$$T(n) = \sum_{i=1}^{n-1} \sum_{j=1}^{n-i} 1$$

(The worst case) is taken

(Here, each loop will do the comparisions)

$$= \sum_{i=1}^{n-1} (n-i)$$

$$= (n-1) + (n-2) + \cdots + (n-n+1)$$

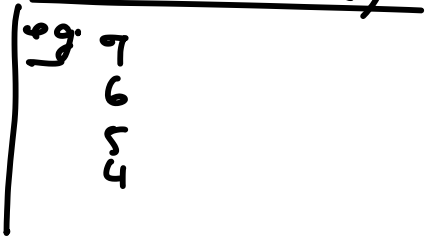
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$$= (n-1) + (n-2) + \cdots + 1$$

$$= \frac{(n-1)\,n}{2} \in \Theta(n^2)$$

Q: What about no of exchanges?

No of exchanges $\in \Theta(n^2)$ (worst case)



Q: Can you make the no of exchanges linear, $\Theta(n)$?